

But nanotechnology will carry this evolution to the next step: deterrence will become obsolete, as it will not be possible to maintain a stable armed peace between nanotechnically-armed rivals. Thus, a runaway nanotechnic arms race may be a race to nowhere; there may be no further island of stable military balance out there, even if we could manage to avoid war along the way.

## War against Terrorism

On March 31 2011, at the 241st National Meeting & Exposition of the American Chemical Society (ACS) in Anaheim, California, Allen Apblett presented a new ink-like explosive detector/neutralizer that can be used anywhere against terrorist explosives including battlefields, airports, and subways. This spray-on material both detects and renders harmless the genre of terrorist explosives responsible for government restrictions on liquids that can be carried onboard airliners.

The material is a type of ink made of tiny metallic oxide nanoparticles. The ink changes color, from dark blue to pale yellow or clear, in the presence of explosives. It also changes from a metallic conductor to a non-conducting material, making electronic sensing possible.

This color-change feature allows the material to work as a sensor for quickly detecting the presence of vapors produced by explosives. Soldiers or firefighters could wear the sensors as badges on their uniforms or use them as paper-based test strips. Airports, subways and other facilities could use the sensors as part of stationary monitoring devices.

The same color-changing material can also serve as an explosives neutralizer. Firefighters and bomb squad technicians could spray the ink onto bombs or suspicious packages until the color change indicates that the devices are no longer a threat. Technicians could also dump the explosives into vats containing the ink to neutralize them.

Peroxide-based explosives, made from hydrogen peroxide, are the most easily made explosive and until now have been very hard to detect. Triacetone triperoxide, or TATP, is another substance which is sometimes used in suicide vests and improvised explosive devices and is not the easiest thing to detect. This new ink provides a quick way to detect and test these explosives, which might be hidden in clothing, food, and beverages. The ink contains nanoparticles of a compound of molybdenum, a metal used in a wide variety of applications including missile and aircraft parts. The dark blue ink reacts with the peroxide explosives and turns yellow or clear. When used as an electronic sensor, the highly-sensitive material is capable of detecting TATP vapors at levels as low as a 50 parts per million, equivalent to a few drops of the vapor in a small room,